

A

Project Stage-I Report

On

**“Enhancing Navigation for Railway
Station Facilities and Locations”**

In partial fulfillment of requirements for the degree of
Bachelor of Technology

In
Computer Science & Engineering (Data Science)

Submitted By

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Computer Science & Engineering (Data Science)

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ABSTRACT

The project SIH-1710: Enhancing Navigation for Railway Station Facilities and Locations aims to transform indoor navigation in complex railway environments through the integration of Augmented Reality (AR) and real-time pathfinding algorithms. By leveraging ARCore, the application dynamically maps indoor layouts without relying on preloaded blueprints, allowing for greater adaptability across different stations. Users can navigate with intuitive AR overlays and directional arrows that update in real time based on their location and destination. A custom pathfinding algorithm computes the shortest, most efficient route through the station, guiding users to platforms, ticket counters, restrooms, and other essential services. Firebase is utilized to store and manage spatial data and user interactions, while the interface is crafted using Figma to ensure a user-friendly experience. This solution addresses key challenges of indoor navigation in high footfall public areas, offering scalable, accessible, and modern assistance to improve commuter convenience and reduce confusion within Indian railway stations.

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CHAPTER- 1

INTRODUCTION

1.1 INTRODUCTION

Navigating large railway stations, especially in a country like India where footfall is immense and infrastructure is vast, can be a daunting task for passengers—particularly those unfamiliar with the premises or facing time constraints.

Traditional signboards and manual assistance often fall short in delivering timely and personalized guidance. To address this challenge, the project SIH-1710: Enhancing Navigation for Railway Station Facilities and Locations presents a cutting-edge solution by integrating Augmented Reality (AR) with real-time indoor navigation.

Using ARCore, the system scans and dynamically maps station interiors, eliminating the need for preloaded architectural blueprints. Users are guided through live AR overlays and directional cues powered by an efficient pathfinding algorithm that calculates the shortest route to any facility—be it platforms, restrooms, ticket counters, or help desks.

With Firebase handling backend data management and a user-friendly UI/UX developed in Figma, this project sets a new benchmark in accessible, scalable, and intelligent indoor navigation solutions.

1.2 PROBLEM STATEMENT

1.2.1 Background:

Railway stations are complex environments with numerous facilities and locations such as ticket counters, platforms, restrooms, food courts, and waiting areas.

Passengers often face difficulties in navigating these spaces, especially in large or unfamiliar stations. Efficient and user-friendly navigation systems are crucial for improving passenger experience, reducing congestion, and ensuring timely travel connections.

1.2.2 Description:

The problem involves developing a comprehensive navigation solution for railway stations that assists passengers in locating various facilities and destinations within the station premises. This includes creating detailed maps, providing real-time directions, and integrating features such as accessibility options for individuals with disabilities. The solution

should be intuitive, easy to use, and accessible via multiple platforms, including mobile devices and digital kiosks.

Key challenges include updating navigation information in real-time, ensuring accuracy, and accommodating the diverse needs of all passengers. Expected Solution: The expected solution is a multi-platform navigation system that provides detailed, real-time directions to all facilities and locations within a railway station.

1.2.3 Facility:

A mobile application with 3D interactive maps and step-by-step navigation. Digital kiosks located throughout the station with touch-screen interfaces.

Voice-guided navigation for visually impaired passengers. Regular updates to reflect changes in station layout and facility locations. Integration with existing railway apps and services for seamless user experience. The solution should enhance the overall passenger experience by reducing confusion, saving time, and improving accessibility within the station

1.3 OBJECTIVES

1 To develop an AR-based indoor navigation system -

Utilize Google ARCore to create a real-time, markerless AR navigation experience that overlays digital directional guidance on the physical environment of railway stations.

2 To eliminate dependency on preloaded blueprints -

Design a dynamic indoor mapping system capable of scanning and tagging key station locations without the need for prior architectural data or fixed floor plans.

3 To implement a reliable pathfinding algorithm -

Integrate a custom pathfinding algorithm that accurately calculates the shortest and most efficient route between the user's current position and selected destinations within the station.

4 To provide a scalable cloud-based data management solution-

Use Firebase to store and manage spatial data, user preferences, real-time updates, and navigation history in a secure and scalable manner.

5 To improve commuter convenience and reduce travel stress -

Minimize the confusion and delays faced by passengers, especially during peak hours or in unfamiliar stations, by offering real-time, on-the-go assistance through their smartphone

1.4 APPLICATION

1. Passenger Navigation in Railway Stations:

The primary application is to assist passengers in navigating through large and complex railway stations using AR overlays, helping them locate platforms, ticket counters, restrooms, waiting rooms, food stalls, and other essential facilities with ease.

2. Assistance for Senior Citizens and Differently-Abled Individuals:

The system provides inclusive navigation support, helping elderly passengers or those with physical challenges move confidently through crowded or unfamiliar station environments without needing constant human assistance.

3. Smart Station Infrastructure Development:

This project can be integrated into smart city initiatives to modernize public transport

infrastructure, making stations more efficient, digitally accessible, and commuter-friendly.

4. Tourist Assistance:

Tourists unfamiliar with local languages or layouts can benefit from the intuitive AR interface, reducing dependency on physical maps or station staff.

5. Emergency Evacuation Guidance:

In case of emergencies, the app can be repurposed to guide users toward the nearest exits or safe zones quickly and safely using real-time route recalculations.

6. Crowd Management and Flow optimization:

Firestore integration, authorities can analyze movement patterns and optimize station layouts or manage crowd flow more effectively.

7. Integration with Other Transport Systems:

The system can be extended to bus terminals, metro stations, or airports, offering a unified navigation experience across multiple transport hubs.

CHAPTER- 2

LITERATURE SURVEY

Indoor navigation has garnered significant attention in recent years due to its potential to transform user experiences in complex environments such as airports, shopping malls, and railway stations. Several research studies have explored various approaches to indoor positioning and AR- guided navigation. Lee et al. (2020) developed a hybrid system combining ARKit and iBeacons, showing high accuracy but facing limitations in scalability due to hardware dependence. In contrast, Chen et al. (2019) proposed a vision-based approach using image anchors, which eliminated the need for beacons and demonstrated promising results in large indoor spaces like malls. Patel et al. (2021) introduced a context-aware pathfinding system using Dijkstra's algorithm, adapting navigation to user preferences and dynamic conditions. Kumar and Sharma (2022) successfully integrated Firebase with ARCore for real-time, cloud-based navigation across university campuses, highlighting the importance of scalable and synchronized data management. Additionally, Park et al. (2018) leveraged SLAM (Simultaneous

Localization and Mapping) for self- updating AR maps, suitable for dynamic environments. These studies collectively emphasize the need for systems that combine flexibility, real- time responsiveness, and minimal infrastructure reliance—principles that directly inform the design and development of the SIH-1710 project.

2.1 BACKGROUND

Railway stations in India are among the busiest and most crowded public areas, serving millions of passengers daily. These stations are often spread across vast areas and house numerous interconnected facilities such as platforms, reservation counters, restrooms, cloakrooms, waiting halls, and food courts. Due to the absence of accessible digital navigation tools, passengers—especially the elderly, tourists, and differently-abled individuals—often experience difficulty in reaching their desired locations within the station in a timely manner. Traditional signage and static maps are limited in their scope and fail to provide dynamic, real-time guidance tailored to individual needs.

To address this gap, emerging technologies such as Augmented Reality (AR), computer vision, and cloud computing offer new possibilities. AR, in particular, has the power to

overlay digital guidance directly onto the user's view of the real world, making navigation intuitive and interactive. Google ARCore, a powerful platform for building AR experiences, enables real-time detection and anchoring of physical surfaces, allowing digital elements such as arrows and markers to be accurately placed in the user's environment. Combined with dynamic pathfinding algorithms and cloud databases like Firebase, such systems can deliver highly responsive and adaptable solutions.

Existing research and commercial systems in this domain have primarily relied on beacon technology, preloaded maps, or image-based recognition. However, these approaches are often either costly to implement on a large scale or lack the flexibility required for frequently changing environments like railway stations. This project, *SIH-1710: Enhancing Navigation for Railway Station Facilities and Locations*, aims to overcome these limitations by developing a scalable, AR-based navigation system that dynamically maps station interiors, computes optimal paths in real time, and provides user-friendly visual guidance—all without the need for static infrastructure or manual intervention. Through this innovative approach, the project

seeks to significantly improve commuter experience, accessibility, and the overall efficiency of railway station navigation.

2.2 EXISTING SYSTEM

Several existing systems and research studies have explored AR and indoor navigation across different domains:

2.2.1 AR Indoor Navigation System using ARKit and iBeacon (Lee et al., 2020)

This study proposed a navigation system that combines Apple's ARKit with Bluetooth beacons (iBeacons) to determine precise indoor positioning. It demonstrated how AR-based directional arrows, anchored via beacon signals, significantly improved user navigation accuracy and experience. However, reliance on physical beacon installation limited its scalability.

Citation:

Lee, J., Kim, S., & Kim, M. (2020). AR Indoor Navigation System Using ARKit and iBeacon. *Sensors*, 20(13), 3754.

<https://doi.org/10.3390/s20133754>

2.2.2 Visual Positioning System for AR Navigation in Large Indoor Spaces (Chen et al., 2019)

This paper introduced a vision-based AR navigation solution that uses image recognition for location estimation. The system eliminated the need for beacons or GPS and proved successful in shopping malls. It emphasized the robustness of image-based anchors and showed potential for application in train stations.

Citation:

Chen, X., Yang, Z., Wang, Y., & Han, J. (2019). Visual Positioning System for Augmented Reality Indoor Navigation. ACM Transactions on Multimedia Computing, Communications, and Applications(TOMM),15(1s),Article26.

<https://doi.org/10.1145/3281450>

2.2.3 SmartPath: A Context-Aware Pathfinding System for Indoor Navigation (Patel et al., 2021)

SmartPath integrates dynamic pathfinding with environmental context awareness (like escalators, crowd density, etc.). It uses Dijkstra's algorithm for shortest path calculation and

adapts paths based on user preferences or physical abilities. The system's modular architecture makes it suitable for integration into larger infrastructure like railway stations.

Citation:

Patel, R., Singh, A., & Nair, K. (2021). SmartPath: A Context-Aware Pathfinding System for Indoor Navigation. *International Journal of Computer Applications*, 183(10), 15–20.

<https://doi.org/10.5120/ijca2021921284>

2.2.4 Firebase-based AR Navigation System (Kumar & Sharma, 2022)

This study implemented an AR navigation system for university campuses using Google ARCore and Firebase for real-time updates. The system's cloud-backed design enabled cross-device data sharing and adaptive mapping, setting a precedent for scalable public navigation tools.

Citation:

Kumar, S., & Sharma, P. (2022). Firebase-Based Augmented Reality Navigation System for Campus Environments. *Journal of Emerging Technologies and Innovative Research (JETIR)*,

9(4), 102–109.

2.2.5 Indoor Navigation using AR and SLAM (Park et al., 2018)

This research focused on combining Simultaneous Localization and Mapping (SLAM) with AR to generate live, self-updating indoor maps. It was especially useful in dynamic environments like malls or airports where layouts change frequently. It concluded that SLAM-based AR can be highly effective in environments where static maps are impractical.

Citation:

Park, H., Kim, D., & Lee, K. (2018). Indoor Navigation System Using AR and SLAM Techniques. IEEE Access, 6, 75547–75559. <https://doi.org/10.1109/ACCESS.2018.2882172>

CHAPTER- 3

METHODOLOGY

3.1 HARDWARE AND SOFTWARE REQUIREMENTS

3.1.1 Hardware Requirements

- **AR-Compatible Smartphones/Tablets:**

Devices that support Google ARCore are essential to render AR overlays and enable real-time scanning of the environment. Minimum specifications include:

1. Android 9.0 (Pie) or above
2. At least 4GB RAM
3. RCore support (camera, motion sensors, depth API)

- **High-Speed Internet Connectivity:**

Required for real-time data retrieval and updates via Firebase. Wi-Fi or 4G/5G internet is necessary to ensure seamless map rendering and path calculation.

- **Development and Testing Devices:**

1. Android Emulator (for non-AR functionality testing)

2. Laptop/PC with at least 8GB RAM and 2.5 GHz processor for software development and debugging

3.1.2 Software Requirements

- **Android Studio:**

The primary Integrated Development Environment (IDE) used for building the Android application. It supports Kotlin and Java, integrates with ARCore SDK, and provides a built-in emulator for testing.

- **Google ARCore SDK:**

Enables the development of Augmented Reality features, including environmental understanding, motion tracking, and placing virtual objects within the real world.

- **Firebase:**

Used as a real-time backend service for:

1. Storing user location data
2. Fetching indoor map updates
3. Managing pathfinding algorithms and user sessions

- **Figma:**

Utilized for UI/UX design and prototyping, ensuring a user- friendly interface for intuitive navigation.

- **Version Control System – Git/GitHub:**

For managing source code, version history, and collaboration among development team members.

- **Pathfinding Algorithm:**

Implemented in the backend or client-side to dynamically calculate the shortest route between the user's current location and their selected destination within the station.

3.2 SYSTEM DESIGN

The system architecture of SIH-1710: Enhancing Navigation for Railway Station Facilities and Locations is bifurcated into two operational modes: Admin Mode for Mapping and User Navigation Mode. This dual-mode design ensures both flexibility and scalability—allowing railway stations to be dynamically mapped by authorized personnel, while delivering real-time AR navigation to end users without dependence on static blueprints.

3.2.1 Mapping the Railway Station (Admin Mode)

The Admin Mode is designed for authorized staff or developers responsible for scanning and tagging key locations within the station. This mode utilizes ARCore's plane detection and anchor placement features to dynamically generate a digital map of the station.

In this process:

- The admin scans the environment using an ARCore-compatible device, identifying physical planes such as floors and walls.
- Virtual anchors are placed at significant locations (e.g., platforms, exits, ticket counters, restrooms).
- These anchors are named, geotagged, and stored in a structured format within Firebase, forming a graph-like map of navigable nodes.
- Each anchor's positional data and metadata are synced in real time to Firebase, ensuring that the layout is available for all users.

This approach removes the need for static floor plans or beacons, making it easy to remap or update the environment as physical station layouts evolve.

3.2.2 User Navigation Mode

The User Navigation Mode activates when a commuter opens the app and selects a destination within the railway station. This mode transforms the device into a live AR navigator.

Key functionalities include:

- The AR camera scans the surroundings and retrieves the pre-mapped anchors from Firebase.
- Based on the user's current position (determined using ARCore's motion tracking), the shortest path to the selected destination is calculated using a pathfinding algorithm such as A*.
- The path is overlaid onto the real-world view via virtual arrows, which adjust dynamically as the user walks.
- At every decision point (e.g., intersections or floor changes), the app provides clear visual cues and optional textual hints to aid the user.

This seamless integration of AR and pathfinding ensures that even first-time visitors or those unfamiliar with the station can navigate confidently and independently.

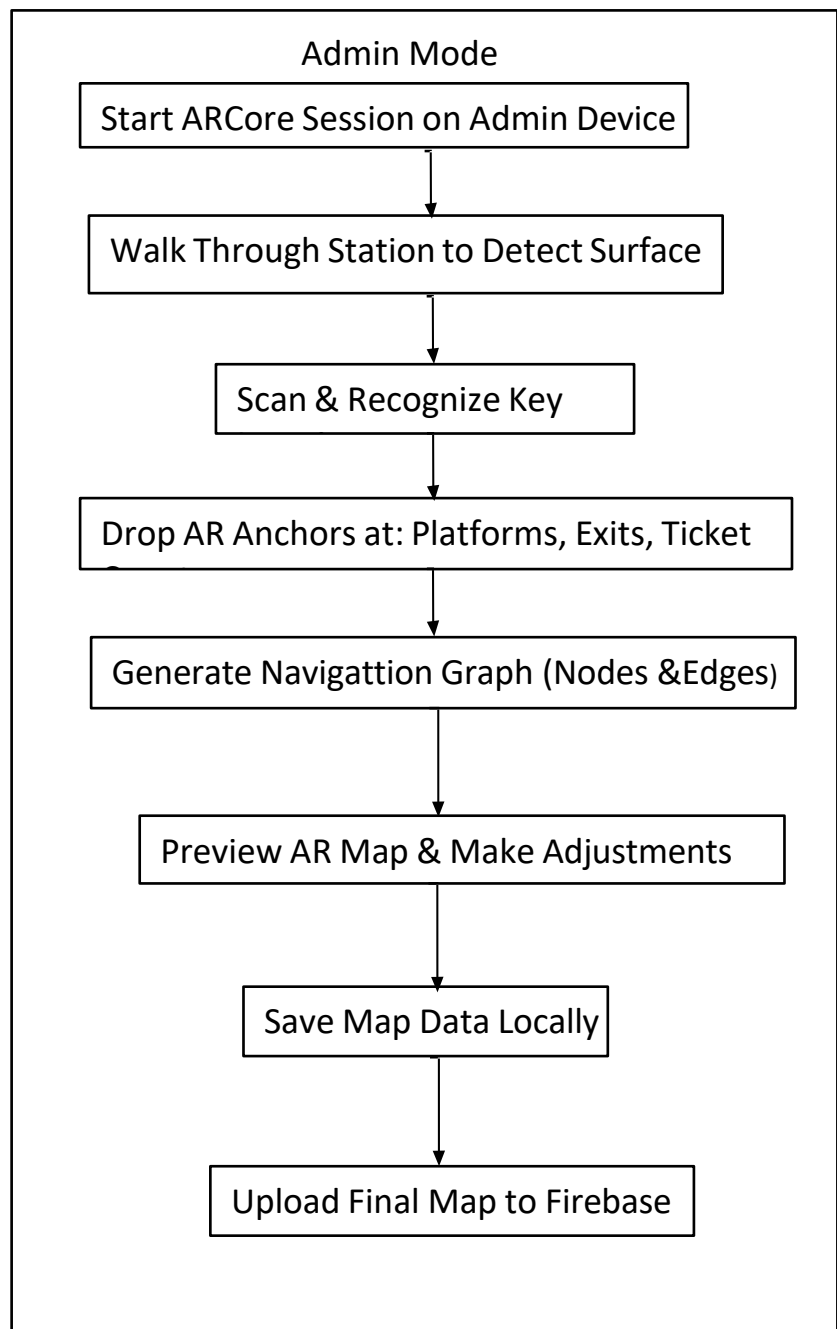
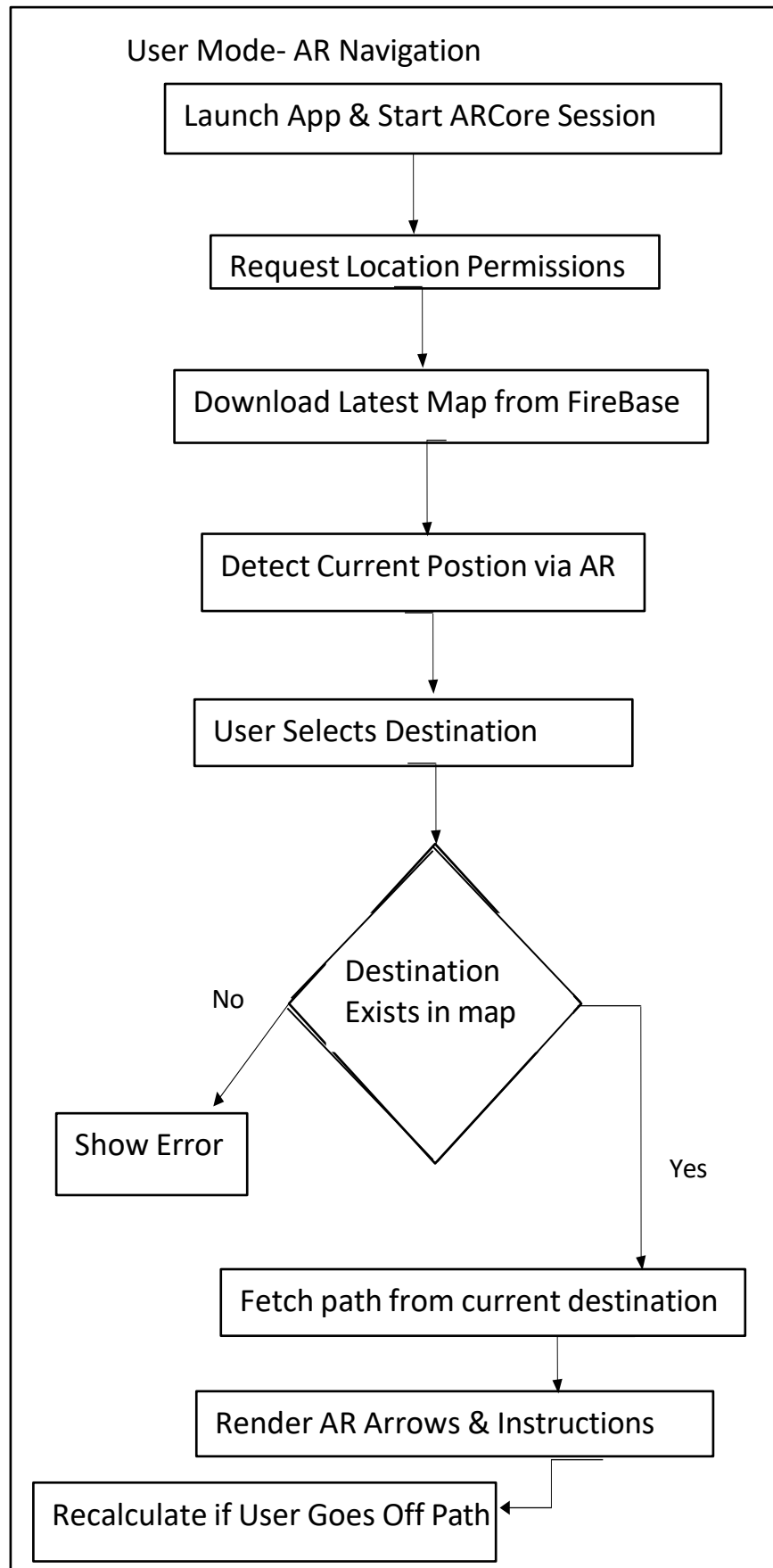


Fig.3.1 Admin Mode Workflow

**Fig.3.2 User Mode Navigation**

3.3 ALGORITHM

To provide efficient, and reliable navigation within the environment of a railway station, the project utilizes the A (A-Star) Pathfinding Algorithm*. This algorithm is widely recognized for its optimal balance between performance and accuracy in finding the shortest path between two points in a graph-based layout.

3.3.1 Why A* Algorithm

The railway station layout, once scanned and mapped by the admin using ARCore, is stored in the form of connected nodes (anchors) representing key locations such as platforms, ticket counters, washrooms, and exits. These nodes form a weighted graph, where:

- **Nodes** represent tagged locations.
- **Edges** represent possible walkable paths between them.
- **Weights** represent the physical distance or time required to traverse that path.

A* is ideal in this case because:

- It combines the benefits of Dijkstra's Algorithm

and Greedy Best-First Search

- It uses a heuristic function ($h(n)$)—commonly the straight-line (Euclidean) distance—to estimate the cost to the goal.
- It minimizes both the actual cost from the start ($g(n)$) and the estimated cost to the goal ($h(n)$) using the evaluation function: $f(n) = g(n) + h(n)$

3.3.2 Working in Our Project Context

1. When the user selects a destination, the algorithm retrieves the users current anchor node and the destination node.
2. The A* algorithm then processes all possible paths through the graph of anchor points, calculating $f(n)$ for each.
3. It prioritizes exploration of the most promising nodes while guaranteeing the shortest valid path.
4. Once the optimal path is determined, it is rendered as a series of AR directional arrows guiding the user step-by-step.

This makes A* the ideal choice for enabling smooth, intelligent, and context-aware indoor navigation in railway stations through our AR-based system.

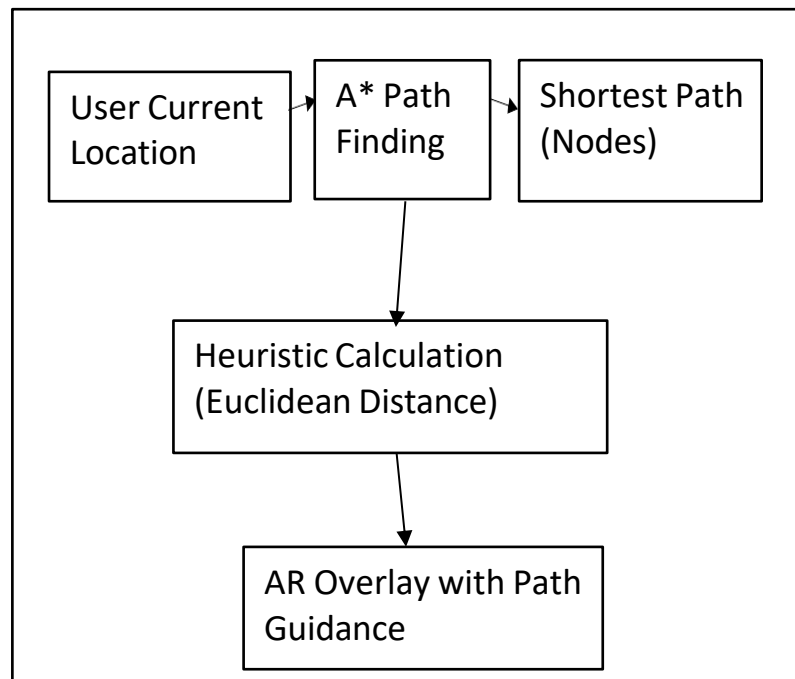


Fig.3.3 Block Diagram of A* Algorithm

CHAPTER-4

IMPLEMENTATION DETAILS

The project has been developed in a modular structure to ensure better maintainability, scalability, and testing. Each module performs a specific function and interacts with others via defined data flows. Below is a detailed module-wise implementation with suggested snapshot stages.

Module 1: Interface Design

Used Figma to design a system that is user-friendly and easy to navigate. We created two portals: one for users and another for admins.

Technologies Used: Figma



Fig4.1: Figma

Module 2: User Interface & Destination Selection

This module allows users to choose a destination from a list (e.g., "Platform 4", "Exit Gate", etc.). The current position is auto-detected using ARCore's motion tracking.

Technologies Used: Android UI (Figma to XML), ARCore

1:05 4G 63%

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Gender
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Date of Birth
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Password
.....

Confirm Password
Re-enter password

 Please fill all fields

CREATE ACCOUNT

Fig. 4. 2: User Interface -Registration

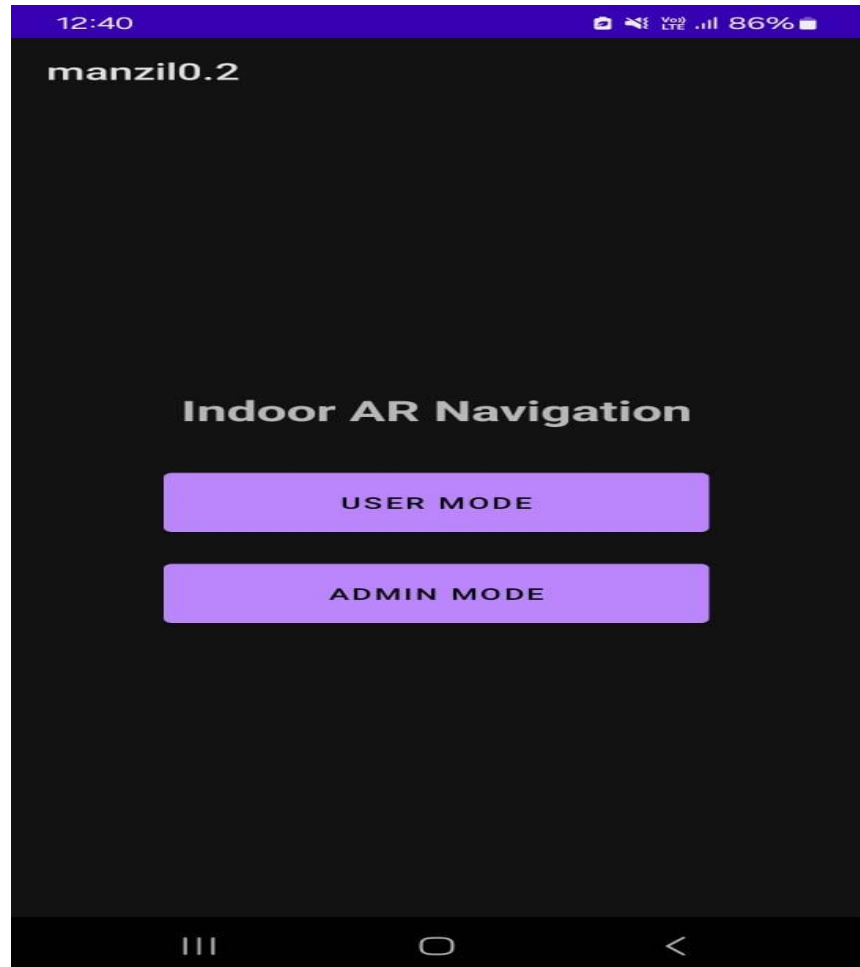


Fig. 4.3: User Interface

Module 3: Admin Mode

This module enables administrators to scan the physical layout of a railway station using ARCore. Anchors (virtual markers) are placed at key points like platforms, ticket counters, restrooms, and exits. Each anchor stores metadata like location name and coordinates.

Technologies Used: Google ARCore, Android Studio, Firebase Database

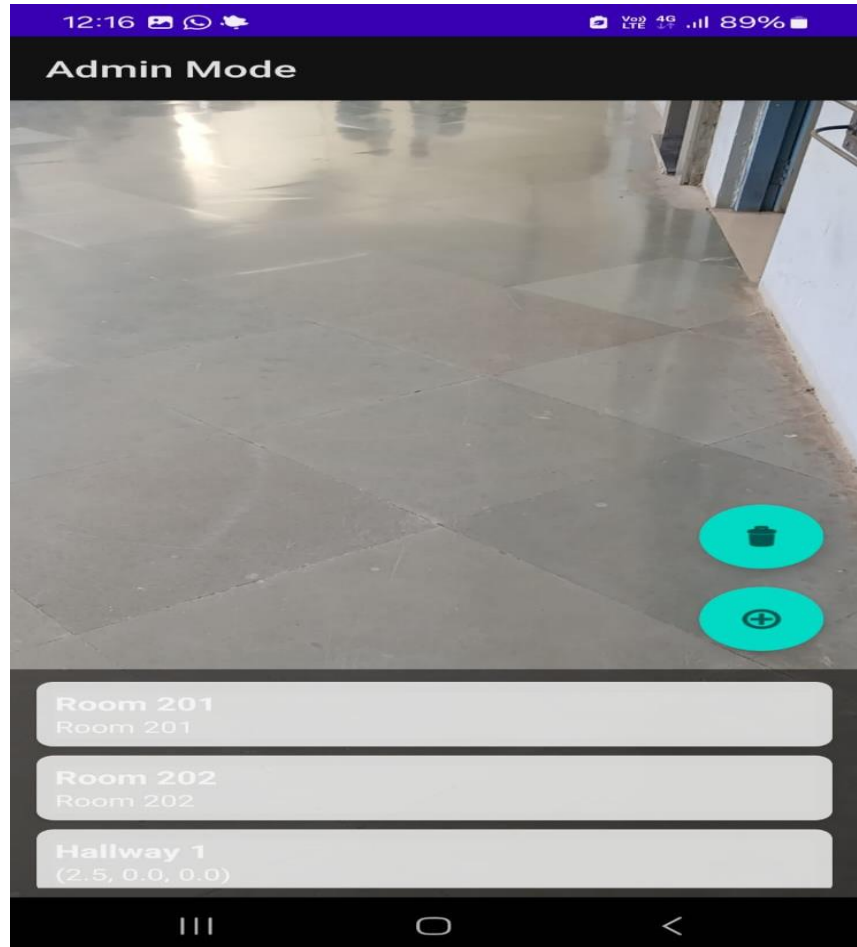


Fig. 4.4: Admin Portal

Module 4: A* Pathfinding Algorithm

Once the current and destination nodes are identified, the A* algorithm computes the shortest path between them using a combination of Euclidean distance and graph traversal. This path is then converted into waypoints for visual guidance.

Technologies Used: Python (initial testing), Java/Kotlin in Android for final implementation

```

private static float heuristic(NavigationNode a, NavigationNode b) {
    float dx = a.getX() - b.getX();
    float dy = a.getY() - b.getY();
    float dz = a.getZ() - b.getZ();
    return (float) Math.sqrt(dx * dx + dy * dy + dz * dz);
}

private static List<NavigationNode> reconstructPath(NodeInfo goal) {
    List<NavigationNode> path = new ArrayList<>();
    NodeInfo current = goal;
    while (current != null) {
        path.add(0, current.node);
        current = current.parent;
    }
    return path;
}

private static NavigationNode findNodeById(List<NavigationNode> nodes, long id) {
    for (NavigationNode node : nodes) {
        if (node.getId() == id) return node;
    }
    return null;
}
}

```

Fig 4.5 Code of A*

CHAPTER -5

RESULT

5.1 DATASET USED

In this project, the dataset is not traditional tabular data but rather a custom-constructed spatial graph dataset. It was created during the admin mapping phase using Google ARCore, which captures the coordinates and metadata of key indoor locations like platforms, entrances, counters, and washrooms.

Each location is saved as a node in Firebase, and logical walkable paths are defined as edges between them. The dataset includes:

1. Anchor ID
2. Coordinates (x, y, z)
3. Location type (platform, counter, etc.)
4. Connected anchors (neighbors)

Since this dataset was manually constructed and dynamically stored using Firebase, there is no third-party dataset citation. However, our approach aligns with real-time mapping methods discussed in:

- [1] Zhou, Q., Wu, Z., & Liu, H. (2021). AR-Based Indoor Positioning System Using Scene Understanding. IEEE Sensors Journal.
- [2] Kim, M. J., et al. (2019). Indoor AR Navigation System Using Pathfinding Algorithms. ACM SIGGRAPH.

CONCLUSION

The development of the project “*SIH-1710: Enhancing Navigation for Railway Station Facilities and Locations*” has progressed significantly up to this stage. Key technical milestones have been achieved, including the successful integration of **Firestore** for real-time data storage and retrieval, the implementation of the **A* pathfinding algorithm** for optimal route calculation, and the complete design and development of a user-friendly **frontend UI/UX** using Figma and Android Studio.

These components have been individually tested and verified for functionality and reliability. However, the final **integration of these modules into a single, unified Android application** is still underway. This step involves combining the real-time AR-based navigation system with pathfinding and data management features to deliver a seamless user experience.

The groundwork laid so far ensures a strong foundation for the remaining work in Semester VII, where full integration, real-world testing, and user feedback will drive the project to completion.

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